Work Trail — Product Requirements Document

1. Product overview

Work Trail is an India-first field-workforce management app for employers and field employees. It records work sessions, periodically captures employee location only during an active work session, lets employees submit typed or voice-to-text work updates, and produces AI-assisted daily, weekly, and monthly summaries with downloadable PDF reports.

Problem

Small and mid-sized businesses with field salespeople, technicians, supervisors, promoters, or service teams often rely on WhatsApp messages, calls, spreadsheets, and verbal reporting. Owners lack a single trustworthy view of: who worked, where they travelled during a shift, what they completed, and what needs follow-up.

Product promise

> Start work, create proof of work, and get a clear report—without making field employees fill long forms.

MVP boundaries

Include: Android-first employee experience; responsive admin web dashboard; roles; invitation and credentials login; work sessions; periodic GPS capture; map trail/history; editable typed or voice-transcribed updates; server-side AI summaries; daily/weekly/monthly reports; PDF export; work-hour calculation.

Exclude from v1: payroll, expense claims, route optimisation, task assignment, chat, facial recognition, covert/background-all-day tracking, geofencing alerts, iOS support, and automatic performance scoring.

2. Users and roles

Role	Primary goal	Permissions
Super admin	Operate one company account	Manage organisation, employees, reports, billing later
Admin/manager	Monitor a team	Invite employees, view assigned/all employee data, download reports
Employee	Prove completed field work quickly	Start/end own work, submit own updates, see own current session

An employee must only access their own data. An admin must only access employees in their organisation. Every API request enforces organisation isolation.

3. Core flows

Admin onboarding

1. Open Work Trail web app.
2. Create company account with company name, admin name, email, mobile number, and password.
3. Verify email/OTP if enabled.
4. Land on empty dashboard with primary CTA: **Invite employee**.
5. Enter employee name, phone/email, designation, and optional employee ID.
6. Send an invitation link by email/SMS/WhatsApp deep link, or create credentials manually.

Employee workday

1. Employee installs Android app and signs in from invitation or assigned credentials.
2. Employee sees status **Off duty** and taps **Start work**.
3. App explains tracking, requests location permissions, creates a work session, and records an immediate location point.
4. While session is active, app captures a location target every five minutes; background delivery is best-effort and visibly indicated through a foreground-service notification.
5. Employee opens **Add update**, types text or speaks into the microphone.
6. Speech becomes editable text. Employee reviews and submits it.
7. Backend stores raw text, generates a structured short AI summary, and links it to the active session.
8. Employee taps **End work**. App records final location, closes session, calculates duration, and stops tracking.

Admin review

1. Admin selects an employee card.
2. Admin opens **Map**, **Updates**, or **Work hours** tabs.
3. Admin chooses day, week, or month period.
4. Dashboard retrieves filtered trail, session totals, updates, and precomputed/on-demand summaries.
5. Admin downloads a PDF report for the selected employee and period.

4. Functional requirements

Authentication and invitations

- Email/password login for admins and employees; mobile OTP can be added after MVP.
- Roles are stored server-side, never trusted from the client.
- Invitation has a single-use, expiring token and is tied to an organisation and intended employee.
- Admin may create an employee account with a temporary password.
- Support forgot-password flow.

Work session

- One active session maximum per employee.
- Start work stores `started_at` using server time and device time, and triggers immediate location capture.
- End work stores `ended_at`, final location attempt, and calculated duration.
- If the app terminates while a session is active, the next launch asks the employee to resume/end the existing session.
- Admin can see active, ended, and incomplete sessions.

Location capture

- Tracking is permitted only when an employee has an active work session.
- Capture every five minutes as the product target, subject to Android OS, network, accuracy, and battery constraints.
- Save latitude, longitude, accuracy, speed if supplied, device timestamp, server receipt timestamp, battery level if available, and source.
- Reject clearly invalid coordinates and retain poor-accuracy records with an accuracy flag rather than silently claiming precision.
- Queue unsent points locally; sync idempotently when network returns.
- Show a persistent Android notification while background tracking is active.
- Do not claim continuous real-time tracking in v1.

Updates and voice input

- Update form has text area, microphone button, category selector (optional), and submit.
- Speech recognition fills the same editable text field; employee must be able to correct it before submission.
- Store raw final text, optional transcript metadata, submission timestamp, and session ID.
- Basic validation: 5–2,000 characters after trimming.

AI summaries

- Generate a concise update summary asynchronously after submission.
- Store raw text separately from generated summary and model metadata.
- Summary generation must not block employee submission.
- Daily reports aggregate updates and session/location facts. Weekly/monthly reports aggregate already-generated daily data where possible.
- Model receives minimal required information; never send passwords, invite tokens, or unrelated employee data.
- Every AI report displays a note: “AI-generated summary based on submitted work updates and recorded work-session data.”

Admin dashboard

- KPI cards: total employees, working now, checked in today, total work hours today, updates submitted today.
- Employee list supports search, status filters, and pagination.
- Employee card: name, designation, current/last status, last update time, today’s work duration.
- Employee profile tabs: Overview, Map, Updates, Work hours, Reports.

Maps

- Date-range presets: Today, Yesterday, This week, Last week, This month, Custom range.
- Map draws chronological point trail as a polyline; start/end markers; point/segment timestamps on inspection.
- Display location confidence using accuracy metadata; do not overstate accuracy.
- Show data gaps where no locations were received for a meaningful period.
- Default map is centred on available employee locations; never expose a map to unauthorised users.

Reports and PDF

- Download PDF per employee and period.
- Report contains company details, employee details, selected range, session table, total hours, update timeline, AI summary, and movement statistics.
- Map image is optional in first implementation; PDF must remain useful without it.
- PDFs are generated server-side, stored with access control, and downloadable only by authorised admins.

5. Screens and routes

Admin web

Route	Screen	Key elements
`/`	Landing	Value proposition, feature preview, login/signup CTAs
`/login`	Login	Email, password, role-aware redirect
`/signup`	Company signup	Company and admin details
`/forgot-password`	Reset password	Email entry and reset confirmation
`/app/dashboard`	Dashboard	KPIs, active employees, recent updates
`/app/employees`	Employees	Search, filters, employee cards, invite CTA
`/app/employees/new`	Invite employee	Invite/create-credentials form
`/app/employees/:id`	Employee profile	Overview, Map, Updates, Work hours, Reports tabs
`/app/reports`	Reports	Generate/download report history
`/app/settings`	Company settings	Company profile, admin profile, retention settings

Employee Android app

Screen	Key elements
Splash/session restore	Authentication check and active-session restore
Login	Credentials login, invite registration link
Permissions	Plain-language location/microphone consent explanation
Home	Status, start/end work CTA, elapsed timer, last sync, add update CTA
Active work	Tracking indicator, session timer, last location time, add update, end work
Add update	Editable textarea, microphone control, optional category, submit
Update submitted	Raw update, AI status/summary when ready
History	Own sessions and updates, read-only
Profile/help	Account, privacy explanation, logout

6. Non-functional requirements

- Android-first. Support Android 10+ initially; test lower-cost devices common in India.
- Secure all APIs with TLS, short-lived access tokens, refresh tokens, and server-side authorisation.
- Store passwords only as strong salted hashes.
- Use UTC in the database; render times in organisation/user timezone, default `Asia/Kolkata`.
- Each write endpoint supports idempotency where offline retries can duplicate data.

- Keep audit logs for invitations, admin access, exports, session changes, and deletion requests.
- Data retention must be configurable per organisation; recommended default: 90 days of precise location data, subject to contractual/legal review.

7. Privacy and trust requirements

- Employee must see a clear disclosure before tracking starts: what is collected, when it is collected, why, who can view it, and how to stop it.
- Tracking occurs only in an active work session, never silently 24/7.
- Show current tracking status and last successful location time in the employee app.
- Require explicit consent suitable for the employer's policies and applicable law. Obtain legal review before commercial launch.
- Use role-based access, organisation isolation, and export audit logs.

8. Success metrics

- At least 80% of active sessions receive a first location point within two minutes of start.
- At least 70% of active workdays contain one or more employee updates.
- At least 90% of submitted updates receive a summary within 60 seconds under normal API conditions.
- Admin can generate a report in under 30 seconds for a typical month.
- Pilot metric: 3 of 5 pilot companies actively use the product for four consecutive weeks.

9. Risks and mitigations

Risk	Mitigation
Android limits background location	Foreground service, clear permission UX, physical-device testing, best-effort wording
Battery drain	5-minute cadence, accuracy filters, stop tracking immediately after end work
Employees reject surveillance	Work-session-only tracking, transparent disclosure, visible status, simple updates that benefit employees
GPS spoofing	Flag suspicious patterns later; do not promise fraud prevention in MVP
AI hallucination	Preserve raw updates, label AI output, use constrained structured prompts, allow admin to view source updates
Weak network	Local queue, retry with idempotency key, sync status in UI

10. MVP acceptance criteria

- Admin can sign up, invite an employee, and view that employee after registration.
- Employee can log in, start/end a session, and see elapsed time.
- At least one Android test device records/syncs location at the five-minute target during an active session.
- Admin sees route points and session hours for selected date ranges.
- Employee can submit an editable typed or speech-derived update.
- Admin sees raw update and AI summary.
- Admin can download a protected PDF report for selected employee/date range.